TEXORA · SESSIONAL NOTES · COMBINED STUDY SHEET

Experiments Based on TE-4105 Industrial Utilities

Three linked practical studies on the utility plant of a textile mill — boiler efficiency by the direct and indirect methods, measurement of humidification performance in a conditioned shed, and a lubrication and maintenance audit of production machinery.

TE-4106 · MAINTENANCE OF TEXTILE MACHINERY SESSIONAL · COMBINED STUDY SHEET · LEVEL 4 TERM 1

1. Aim and organisation of the studies

Aim: to measure, calculate and judge the performance of the three utility systems on which every textile process depends — steam generation, air conditioning (humidification), and the lubrication/maintenance system that keeps production machines available. TE-4105 studied these systems in theory; this sessional turns the theory into measurements. Each study follows the same discipline: state what is measured and with what instrument, record honestly, calculate step by step, and compare the result with the benchmark an engineer would expect.

STUDY	SYSTEM	KEY INSTRUMENTS	PERFORMANCE MEASURE
A	Boiler house	Feed-water meter, coal weighbridge records, steam tables, flue-gas thermometer	Thermal efficiency (%) — direct and indirect methods
B	Humidification plant and shed air	Sling (whirling) psychrometer, dry- and wet-bulb thermometers	RH, humidity ratio, water evaporation duty (kg/h)
C	Production machinery	Oil issue records, viscometer, breakdown logbook	Oil consumption, MTBF, MTTR, availability (%)

2. Study A — Boiler efficiency test

2.1 Theory

A boiler's **thermal efficiency** is the fraction of the fuel's heat that actually leaves in the steam. The **direct method** compares heat absorbed by steam with heat supplied by fuel over the same period:

$$\eta = [\, m_s \times (h_s - h_{fw}) \,] \div [\, m_f \times \mathrm{GCV} \,] \times 100\ \%$$

where m_s is steam raised (kg), h_s the enthalpy of steam at the working pressure, h_{fw} the enthalpy of feed water, m_f the fuel burned (kg) and GCV the gross calorific value of the fuel (kJ/kg). The **indirect method** instead measures the losses — dry flue gas, moisture in fuel and hydrogen, incomplete combustion, radiation, blowdown — and subtracts them from 100 %. The two methods should agree within a few per cent; a large disagreement means a metering error, and finding it is part of the exercise.

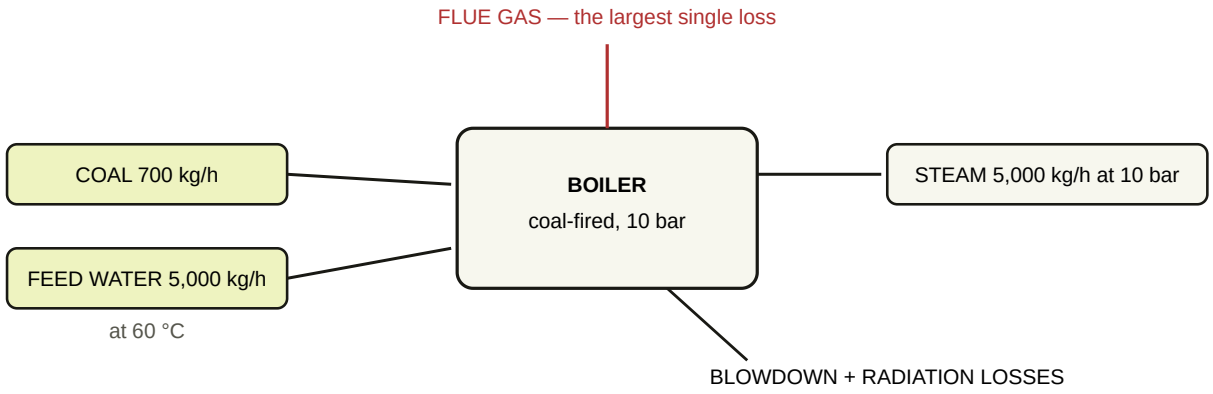


Fig. 1 — Boiler heat-balance boundary for Study A: everything crossing the boundary is metered or estimated.

2.2 Recorded data and worked calculation

Steady one-hour test on the mill's coal-fired boiler: steam raised 5,000 kg/h at 10 bar (saturated, h_s = 2,778 kJ/kg from the steam tables); feed water at 60 °C (h_{fw} = 251 kJ/kg); coal burned 700 kg/h, GCV 25,000 kJ/kg.

> **WORKED CALCULATION — DIRECT METHOD**
>
> Heat absorbed by steam = 5,000 × (2,778 − 251) = 5,000 × 2,527 = **12,635,000 kJ/h**.
>
> Heat supplied by fuel = 700 × 25,000 = **17,500,000 kJ/h**.
>
> η = 12,635,000 ÷ 17,500,000 × 100 = **72.2 %**.

Indirect method (loss account, % of fuel heat): dry flue gas 8.5; moisture in fuel and from hydrogen 4.2; incomplete combustion 1.5; radiation and convection 2.0; blowdown 1.8; unaccounted (balance) 9.8 — total losses 27.8 %, so η = 100 − 27.8 = **72.2 %**. The two methods agree, which validates the metering. The lessons: flue gas is the dominant controllable loss (excess air and stack temperature are the levers — every 20 °C of unnecessary stack temperature costs roughly 1 % of efficiency), and the 9.8 % unaccounted balance warns that ash carbon and intermittent blowdown were only estimated.

3. Study B — Humidification measurement

3.1 Theory

Textile sheds are conditioned because fibre behaviour — regain, strength, static, breakage rate — follows the air's **relative humidity (RH)**. The field instrument is the **sling psychrometer**: two thermometers, one dry-bulb (DBT) and one with a wet muslin sleeve (WBT). Evaporation from the wet sleeve cools it; the drier the air, the larger the depression (DBT − WBT). From the two readings the actual vapour pressure follows from the psychrometric relation:

$$p_v = p_{ws}(\mathrm{WBT}) - 6.6 \times 10^{-4} \times P \times (\,\mathrm{DBT} - \mathrm{WBT}\,)$$

$$\mathrm{RH} = p_v \div p_{ws}(\mathrm{DBT}) \times 100\ \% \quad | \quad W = 0.622 \times p_v \div (\,P - p_v\,)$$

where $p_{ws}(T)$ is the saturation vapour pressure at temperature T, P is atmospheric pressure (101.325 kPa) and W is the **humidity ratio** (kg water per kg dry air) — the quantity that sizes the humidification plant, because the plant must supply the difference between the target W and the incoming air's W for every kilogram of air circulated.

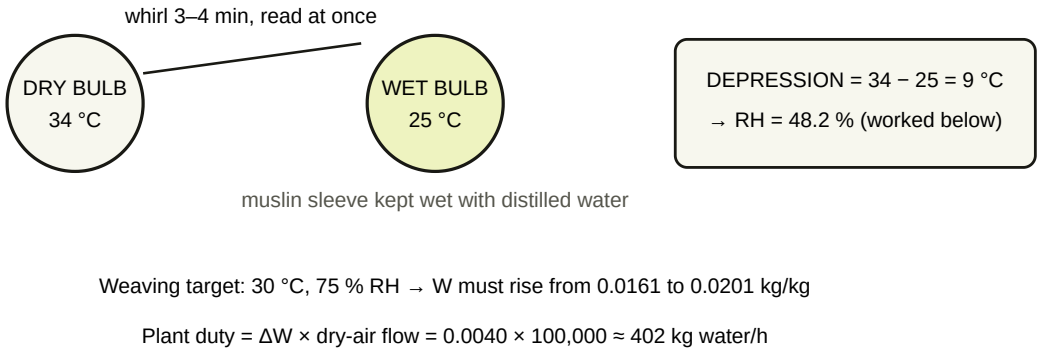


Fig. 2 — Sling psychrometer readings for Study B and the duty they imply for the humidification plant.

3.2 Recorded data and worked calculation

Shed readings (average of three positions): DBT 34 °C, WBT 25 °C. Saturation pressures from the tables/psychrometric chart: $p_{ws}(25\ °C) = 3.162$ kPa; $p_{ws}(34\ °C) = 5.314$ kPa; $p_{ws}(30\ °C) = 4.237$ kPa.

WORKED CALCULATION — RH, HUMIDITY RATIO AND PLANT DUTY

$p_v = 3.162 - 6.6 \times 10^{-4} \times 101.325 \times (34 - 25) = 3.162 - 0.602 =$ **2.560 kPa**.

RH = 2.560 ÷ 5.314 × 100 = **48.2 %** — far too dry for weaving.

W (shed now) = 0.622 × 2.560 ÷ (101.325 − 2.560) = **0.0161 kg/kg dry air**.

Target 30 °C / 75 % RH: $p_v = 0.75 \times 4.237 = 3.178$ kPa; W = 0.622 × 3.178 ÷ (101.325 − 3.178) = **0.0201 kg/kg**.

ΔW = 0.0201 − 0.0161 = 0.0040 kg/kg. At a circulated dry-air flow of 100,000 kg/h, evaporation

duty = 0.0040 × 100,000 ≈ **402 kg of water per hour** — the figure the atomiser/air-washer plant must deliver, before allowing for fresh-air intake and leakage.

The study also records *where* readings were taken: RH varies across a shed — near doors and hot machine rows it falls — so control sensors belong at representative working positions, not beside the plant room outlet.

4. Study C — Lubrication and maintenance audit

4.1 Theory

Lubrication fails quietly: a starved bearing does not stop the mill today, it stops it next month. The audit therefore works from records — oil issued per machine per period, oil condition (colour, grit, viscosity change), and the breakdown log — and converts them into the standard reliability measures:

$$\text{MTBF} = \text{total running time} \div \text{number of breakdowns} \quad | \quad \text{MTTR} = \text{total repair time} \div \text{number of breakdowns}$$

$$\text{Availability} = \text{MTBF} \div (\text{MTBF} + \text{MTTR}) \times 100\ \%$$

4.2 Recorded data and worked calculation

Oil consumption — ring frames: 1,200 spindles audited; spindle-bolster top-up averages 0.5 mL per spindle per week → 1,200 × 0.5 = 600 mL per week = 0.6 L/week, i.e. **31.2 L per year** for the frame set — the baseline against which a sudden rise (a leak or over-oiling) would stand out. Oil found dark and gritty at two frames traced to a missing filler-cap felt: a lubrication fault that is really a housekeeping fault.

WORKED CALCULATION — RELIABILITY OF A SIZING MACHINE

One month: scheduled 720 h; 6 breakdowns; total repair time 18 h; running time 720 − 18 = 702 h.

MTBF = 702 ÷ 6 = **117 h**. MTTR = 18 ÷ 6 = **3.0 h**.

Availability = 117 ÷ (117 + 3) × 100 = **97.5 %**.

Interpretation: the machine fails about twice a week but is repaired quickly; improvement should target failure *causes* (the log shows 4 of 6 stops from the same size-box steam trap), not repair speed.

The audit closes with the maintenance system itself: is the schedule preventive (calendar/task based), predictive (condition based) or still breakdown-driven? Records from this study — one repeated root

cause producing two-thirds of stops — are exactly the evidence that moves a mill from breakdown maintenance toward planned maintenance, the subject developed fully in TE-4105 and TE-4107.

5. Precautions

- **Boiler:** run the efficiency test only at steady load for a full hour; log coal and water over identical start/stop meter readings; never open a blowdown valve during a timed test without recording it.
- **Psychrometry:** sleeve must be clean muslin wetted with distilled water; whirl until two successive readings agree; shield bulbs from radiant heat (a reading beside a hot size box is a local reading, not a shed reading).
- **Audit:** reconcile oil issued with oil purchased and oil in store — issue records alone can hide wastage; wear gloves handling used oil and dispose of it through the mill's waste-oil channel, never to drain.
- All pressure work follows the boiler attendant's instructions; students observe and record — they do not operate boiler mountings.

6. Practice (viva preparation)

1. A boiler raises 4,000 kg/h of steam ($h_s - h_{fw}$ = 2,450 kJ/kg) burning 560 kg/h of coal of GCV 24,500 kJ/kg. Find the efficiency by the direct method.
2. Why do the direct and indirect boiler methods rarely agree exactly, and which discrepancies point to metering faults?
3. DBT 32 °C, WBT 26 °C. Using p_{ws}(26 °C) = 3.363 kPa and p_{ws}(32 °C) = 4.759 kPa, compute RH. (Psychrometric constant as in this sheet.)
4. A shed target is 27 °C / 65 % RH. Explain why chasing the target by RH alone, without humidity ratio, misleads the plant operator when outside temperature changes.
5. A machine logs 5 breakdowns and 12 h repair in 600 scheduled hours. Compute MTBF, MTTR and availability.
6. State two ways a lubrication audit detects a fault that has not yet caused a breakdown.

Answer key

1. Heat to steam = 4,000 × 2,450 = 9,800,000 kJ/h. Fuel heat = 560 × 24,500 = 13,720,000 kJ/h. η = 9,800,000 ÷ 13,720,000 × 100 ≈ 71.4 %.
2. The direct method depends on accurate steam, feed-water and fuel metering over exactly the same interval, while the indirect method estimates losses that are hard to measure (radiation, unburnt carbon, unaccounted balance). Unburnt carbon in ash and intermittent blowdown are usually

estimated, so a few per cent of disagreement is normal; a large gap — say the direct figure far below the indirect — usually means unmetered steam use, a faulty feed meter or coal weighed over a different period than the water.

3. $p_v = 3.363 - 6.6 \times 10^{-4} \times 101.325 \times 6 = 3.363 - 0.401 = 2.962$ kPa. RH = 2.962 ÷ 4.759 × 100 ≈ 62.2 %.
4. RH depends on temperature as well as moisture content: the same air at the same moisture content shows a higher RH when cooled and a lower RH when heated. Humidity ratio W (kg water per kg dry air) measures the moisture itself. Plant duty — how much water to evaporate — is fixed by the change in W and the air mass flow, so operators and controls should track W (or dew point), using RH as the comfort/process indicator at the working temperature.
5. Running time = 600 − 12 = 588 h. MTBF = 588 ÷ 5 = 117.6 h. MTTR = 12 ÷ 5 = 2.4 h. Availability = 117.6 ÷ (117.6 + 2.4) × 100 = 98.0 %.
6. By consumption drift — a machine or frame set suddenly using more oil than its baseline (leak, worn seals, over-oiling) or less (blocked feed lines starving bearings); and by oil condition — darkening, grit, metal particles or viscosity change in drained oil reveals wear in progress long before failure, especially when trended sample by sample.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib